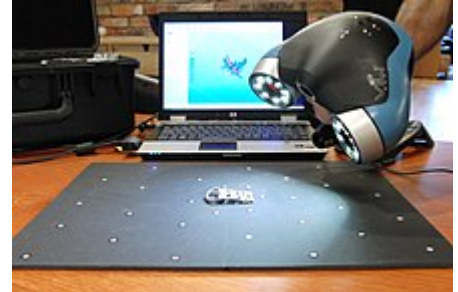


3D scanning

3D scanning is the process of analyzing a real-world object or environment to collect data on its shape and possibly its appearance (e.g. colour). The collected data can then be used to construct digital 3D models.

A **3D scanner** can be based on many different technologies, each with its own limitations, advantages and costs. Many limitations in the kind of objects that can be digitised are still present. For example, optical technology may encounter many difficulties with shiny, reflective or transparent objects. For example, industrial computed tomography scanning and structured-light 3D scanners can be used to construct digital 3D models, without destructive testing.

Collected 3D data is useful for a wide variety of applications. These devices are used extensively by the entertainment industry in the production of movies and video games, including virtual reality. Other common applications of this technology include augmented reality,^[1] motion capture,^{[2][3]} gesture recognition,^[4] robotic mapping,^[5] industrial design, orthotics and prosthetics,^[6] reverse engineering and prototyping, quality control/inspection and the digitization of cultural artifacts.^[7]



Making a 3D-model of a Viking belt buckle using a hand held VIUScan 3D laser scanner.

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Functionality

Technology

- Contact
- Non-contact active
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 - User assisted (image-based modelling)

Reconstruction

- From point clouds
- From models
- From a set of 2D slices
- From laser scans

Applications

- Construction industry and civil engineering